

Loss / objective function variants

Question: is β_{NLL} (Seitzer 2022) a better regression objective than standard Gaussian NLL?

Expectation:

- β_{NLL} @ $\beta=0$ $\rightarrow$ pure NLL.
- $\beta > 0$ reweights per-sample loss by variance^{β} — stabilises training when uncertainty is very non-uniform.
- On heteroscedastic toys we expect a small β_{NLL} improvement; on homoscedastic toys it should match NLL.

ProbReg: best per (dataset, loss, beta)

dataset	loss	beta	mse	nll	crps
b1_gravitationa	nll	0.500	0.066	-0.477	0.116
bimodal	nll	0.500	2.328	1.845	0.909
heteroscedastic	nll	0.500	1.592	1.352	0.635

Multi-seed mean +/- std on heteroscedastic

model	n_seeds	mse_mean	mse_std	nll_mean	nll_std
FlexNN(ELBO)	5	1.3876	0.0244	1.2771	0.0157
LightGBM	5	1.3791	0.0536	1.5802	0.0194
ProbReg(k=3)	5	2.0966	0.3773	1.5472	0.1219
PyTorchNN	5	1.4365	0.1019	1.3662	0.1060
XGBoost	5	1.4923	0.1040	1.6549	0.0281

Verdict — loss functions

Verdict: NOT FULLY ANSWERED: the subsampled ablation used NLL only (beta_NLL cel